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Case Fatality of Subarachnoid Hemorrhage by Aneurysm Location

A Population-Based Study From Finland and New Zealand

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Abstract

Background and Objectives

According to region-specific hospital-based studies, the location of ruptured intracranial aneurysm (RIA) influences the case fatality rate (CFR) of aneurysmal subarachnoid hemorrhage (SAH). However, little is known about whether CFRs vary by RIA location in population-based studies that include prehospital SAH deaths. We assessed whether CFRs and CFR-trends differ by RIA location using whole population data from Finland and New Zealand.

Methods

We used externally validated administrative databases to identify all nonhospitalized and hospitalized SAH cases in Finland and New Zealand from 2001 to 2017. Using the ICD-10, we categorized RIAs into anterior communicating artery (Acom) (ICD-10 I60.2), internal carotid artery (ICA) (I60.0/I60.3), middle cerebral artery (MCA) (I60.1), and vertebrobasilar artery (VBA) (I60.4/I60.5) locations. To validate the location-specific SAH diagnoses, we used external hospital- and population-based datasets and autopsy data. We calculated sudden death rates (those occurring before admission to a ward) and overall 30-day CFRs, and computed age-, sex, and country-adjusted risk ratios using a Poisson regression model with 95% CIs.

Results

Among 13,470 SAH cases (5,056 from New Zealand; median age 58 years; 61.3% women), 26.6% had Acom, 18.4% ICA, 29.5% MCA, 11.5% VBA, and 14.0% other/unspecified RIAs. The overall 30-day CFRs were the greatest for VBA (54.1%), followed by MCA (40.5%), Acom (29.1%), and ICA (28.5%) RIAs. Location-specific sudden death rates were 33.0%, 21.6%, 11.9%, and 9.9%, respectively. Between 2001–2003 and 2015–2017, overall 30-day CFRs declined significantly for VBA (24%, 95% CI 13%–34%) and MCA (15%, 95% CI 5%–24%) RIAs. Location-specific differences in CFRs were similar between countries, but temporal decreases were observed only in Finland. Between 2001–2003 and 2014–2017, the proportion of VBA RIAs increased by 35.9% (from 10.3% to 14.0%).

Discussion

SAH CFRs vary significantly by RIA location, with VBA and MCA RIAs having the greatest CFRs, mainly due to high sudden death rates. It is unclear whether aneurysms with these high-risk locations could benefit from improved primary/secondary prevention or prehospital management strategies. These findings cannot be directly applied to the preventive treatment of unruptured intracranial aneurysms.

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Glossary

Acom = anterior communicating artery; **ARCOS** = Auckland Regional Community Stroke Studies; **BA** = basilar artery; **CFR** = case fatality rate; **CRHC** = Care Register for Health Care; **HUH** = Helsinki University Hospital; **ICA** = internal carotid artery; **ICD-10** = International Classification of Diseases, 10th Revision; **ICH** = intracerebral hemorrhage; **IVH** = intraventricular hemorrhage; **KUH** = Kuopio University Hospital; **MCA** = middle cerebral artery; **NMDS** = National Minimum Dataset; **NZ** = New Zealand; **Pcom** = posterior communicating artery; **PPV** = positive predictive value; **RIA** = ruptured intracranial aneurysm; **SAH** = subarachnoid hemorrhage; **TAUH** = Tampere University Hospital; **VA** = vertebral artery; **VBA** = vertebrobasilar artery; **WH** = Wellington Hospital.

Introduction

Although the incidence of aneurysmal subarachnoid hemorrhage (SAH) has declined worldwide,[1] case fatality rates (CFRs) have decreased only modestly,[2] keeping SAH among the leading causes of cardiovascular and neurologic mortality.[1] Determining SAH CFRs and associated risk factors is challenging as >50% of deaths occur prehospitalization, making hospital-based studies biased.[3,4] In addition, most population-based studies combine SAH cases due to low number of events, despite notable differences in treatment approaches, complication rates, and prognosis. For example, although hospital-based studies provide evidence that SAH CFRs vary considerably by location of a ruptured intracranial aneurysm (RIA),[5-9] only 2 small population-based studies that included prehospital deaths have analyzed SAH CFRs by RIA location, and both collected region-specific data over 4 decades ago.[10,11] Therefore, robust and current evidence on location-specific SAH CFRs and especially prehospital sudden death rates is lacking.

To assess whether SAH CFRs differ by RIA location, we conducted an international population-based study using both nonhospitalized and hospitalized SAH cases in Finland and New Zealand (NZ) from 2001 to 2017. Based on previous hospital-based studies,[5-9,12] we hypothesized that prehospital and 30-day SAH CFRs vary by RIA location, and that the proportion of high-CFR locations (e.g., posterior circulation) has increased during the early 21st century.

Methods

Standard Protocol Approvals, Registrations, and Patient Consents

This study received approval from the Helsinki University Hospital (HUH) institutional review board (HUS/182/2021), the Finnish Institute for Health and Welfare (THL/3136/14.06.00/2021), Statistics Finland (TK/2920/07.03.00/2023), the NZ Ministry of Health (2022-2312), and the Northern Health and Disability Ethics Committee of New Zealand (2023-AM 9094). Finnish and NZ legislations exempt register-based studies using pseudonymized data from obtaining informed consent. This study followed the Strengthening the Reporting of Observational Studies in Epidemiology guideline.

Case Ascertainment

Register-based case ascertainment protocols are described in previous studies and in the eMethods.[13,14] Briefly, both cohorts were derived from nationwide administrative databases, including data on hospital discharges and causes of death. Using ICD-10 codes I60.0–I60.9, the hospital discharge collections (i.e., Care Register for Health Care [CRHC] in Finland[13] and National Minimum Dataset [NMDS] in New Zealand),[14] we captured all publicly funded hospital discharges where SAH was registered as the primary or secondary diagnosis (eFigure 1). Because the Finnish data covered the years 1998–2017 and the NZ dataset 2001–2018, we focused on the overlapping period of 2001–2017. We used data from the Cause of Death Register in Finland and the Mortality Collection in New Zealand to identify SAH deaths occurring prehospital and posthospital admission. The autopsy rate for patients dying outside hospitals is high in both populations; in New Zealand, the autopsy rate is approximately 80% among individuals <40 years and 50% among those aged 40–60 years, whereas in Finland, the corresponding rate for individuals aged 40–64 years is 87%.[14,15] To achieve accurate case identification, the data underwent external case validation.[16] In Finland, we compared administrative cases admitted to HUH neurosurgery with manually maintained hospital-based SAH registers (HUH Intracranial Aneurysm Register 1998–2008[12] and HUH Aneurysm Repair Register 2012–2017[17]), where all cases were verified through CT angiography, digital subtraction angiography, or surgery. In New Zealand, the administrative registration data were compared against SAH cases identified from the Auckland Regional Community Stroke Studies (ARCOS) III and IV,[18,19] which are prospective population-based incidence cohorts in the Greater Auckland region encompassing around one-third of the NZ population. After systematically excluding false positive registrations (e.g., misclassified and non-aneurysmal SAHs) from administrative collections, both cohorts demonstrated high case identification accuracy. For patients admitted to neurosurgical units, a 98.6% positive predictive value (PPV) and 91.3% sensitivity was observed for SAH diagnosis in the Finnish dataset, and a 95.8% PPV with 95.0% sensitivity in the NZ dataset (eTables 1–4).

External Validation of RIA Locations

We also validated the ICD-10 registration codes for more specific RIA locations of included cases. For information on this, see eMethods (pp. 3–5). Briefly, the validation was primarily based on definitions of 6 location-specific ICD-10 codes for SAH from the following: (1) carotid siphon and bifurcation (I60.0), (2) middle cerebral artery (MCA) (I60.1), (3) the anterior communicating artery (Acom) (I60.2), (4) the posterior communicating artery (Pcom) (I60.3), (5) the basilar artery (BA) (I60.4), and (6) the vertebral artery (VA) (I60.5). However, consistent with the standard categorization used in many clinical practices and trials,[5,6,9] we performed the final validation and case fatality analyses in 4 different categories: (1) ICA RIAs (I60.0 and I60.3), (2) MCA RIAs (I60.1), (3) Acom RIAs (I60.2), and (4) vertebrobasilar artery (VBA) RIAs (I60.4 and I60.5). This produced the most accurate validation results (eMethods, pp. 3–5). Regarding cases without location-specific ICD-10 codes, we excluded "SAHs from other intracranial arteries" (I60.6), "SAHs from unspecified intracranial arteries" (I60.7), "other SAHs" (I60.8), and "unspecified SAHs" (I60.9) from the main CFR analyses. To validate location-specific SAH diagnosis among hospitalized patients, we compared patients identified from the HUH Aneurysm Repair Register[17] and ARCOS IV[19] with cases identified via administrative datasets and calculated the proportion of patients with correctly coded ICD-10 diagnoses (i.e., location-specific PPV and sensitivity) (eTables 5–7). Because the Finnish HUH Aneurysm Repair Register did not include conservatively treated patients with SAH and CRHC did not have data based on which patients with SAH underwent operative treatment, we were not able to calculate location-specific PPVs for Finnish patients. However, we cross-checked the region-specific distribution of RIA locations in the Finnish CRHC against those reported in previously published hospital-based SAH cohorts from Tampere University Hospital (TAUH)[20] and Kuopio University Hospital (KUH) (eTable 8).[21] Similarly, we compared the NZ location-specific administrative registrations with previously reported estimates in the neurosurgical department of Wellington Hospital (WH) (eTable 8).[22] Finally, location-specific accuracy of sudden-death SAHs recorded in the cause-of-death collections reflected the proportion of cases that underwent autopsy. In both countries, an investigation of the source and location of the bleeding is routinely performed whenever SAH is observed during autopsy (as per guidelines[23]), thus providing accurate location-specific causes of death.

Demographic Data and Outcomes

We extracted age, sex, municipality of residence, date of SAH, and 365-day survival for all SAH cases in both datasets. Consistent with prior research,[3] we classified SAH cases into young (<40 years), middle-aged (40–64 years), and older (≥65 years) patients. In addition, we gathered information on treatment hospitals and their specialties for hospitalized patients, and the place of death and postmortem examinations for fatal SAH cases. For NZ data, we grouped cases by ethnicity in the following order of prioritization if multiple ethnicities were recorded for a single case: Māori, Pacific, Asian (East, South, and Southeast Asian), and European/others.[24] We did not have ethnicity data for Finnish patients, but it is estimated that ∼2% were of non-European descent. We categorized patients with SAH by municipality of residence into 21 hospital districts in Finland and 20 in New Zealand, encompassing all geographical areas of both countries (eFigure 2). We had no information about treatment modalities, aneurysm size, or comorbidities. The primary outcome measure was overall 30-day CFR of SAH, regardless of underlying cause of death. Furthermore, we measured the proportion of prehospital SAH deaths and the 30-day CFRs among hospitalized SAH patients. Similar to previous administrative studies,[3,14] prehospital sudden deaths were identified via cause-of-death collections but without a SAH-related hospitalization period.

Statistical Analysis

To compare baseline characteristics between subgroups, we applied χ^2 tests to categorical and Mann-Whitney *U* tests to continuous variables. We calculated crude prehospital sudden death rates and 30-day CFRs (overall and after hospitalization) by RIA location (Acom, ICA, MCA, and VBA). Given RIA locations may vary by sex,[25] age,[25] ethnicity,[22] and geographical region, we analyzed case fatalities by these groups separately. We used Poisson regression to calculate age-, sex-, and country-adjusted risk ratios with 95% CIs for fatal SAH by RIA location. We made no adjustments for RIA treatment modalities due to lack of data. Because of the largest case number, MCA RIAs served as a reference category in all analyses. To examine temporal trends in case fatality by RIA locations, we calculated age-, sex-, and country-adjusted average annual percent changes for CFRs using Poisson regression. We further compared 3-year average rates for the first (2001–2003) and last 3 years (2014–2017) of the study period. We used Stata version 17.0 (StataCorp., College Station, TX) for all statistical analyses.

Data Availability

The dataset is unavailable for public use due to data privacy restrictions. However, qualified researchers can apply for access through the Finnish Health and Social Data Permit Authority and NZ Ministry of Health.

Results

Accuracy of the Register-Based RIA Localization

The location-specific sensitivity of the Finnish CRHC ranged between 80.4% for ICA and 86.0% for VBA RIAs (eTable 5). The NZ NMDS demonstrated a location-specific sensitivity between 75.0% for VBA and 95.2% for Acom RIAs, whereas PPV ranged from 72.7% for MCA to 100.0% for VBA RIAs (eTables 6 and 7). Hospitalized patients with SAH in the Finnish CRHC and NZ NMDS had similar region-specific distributions of RIA locations to their relevant published SAH cohorts (TAUH and KUH in Finland, and WH in New Zealand) (eTable 8). Autopsies for prehospital SAH deaths were 90.8% (91.4% in Finland and 87.1% in New Zealand), being relatively similar by age, sex, ethnicity, study year, geographical region, and RIA location (eTable 9).

General Characteristics

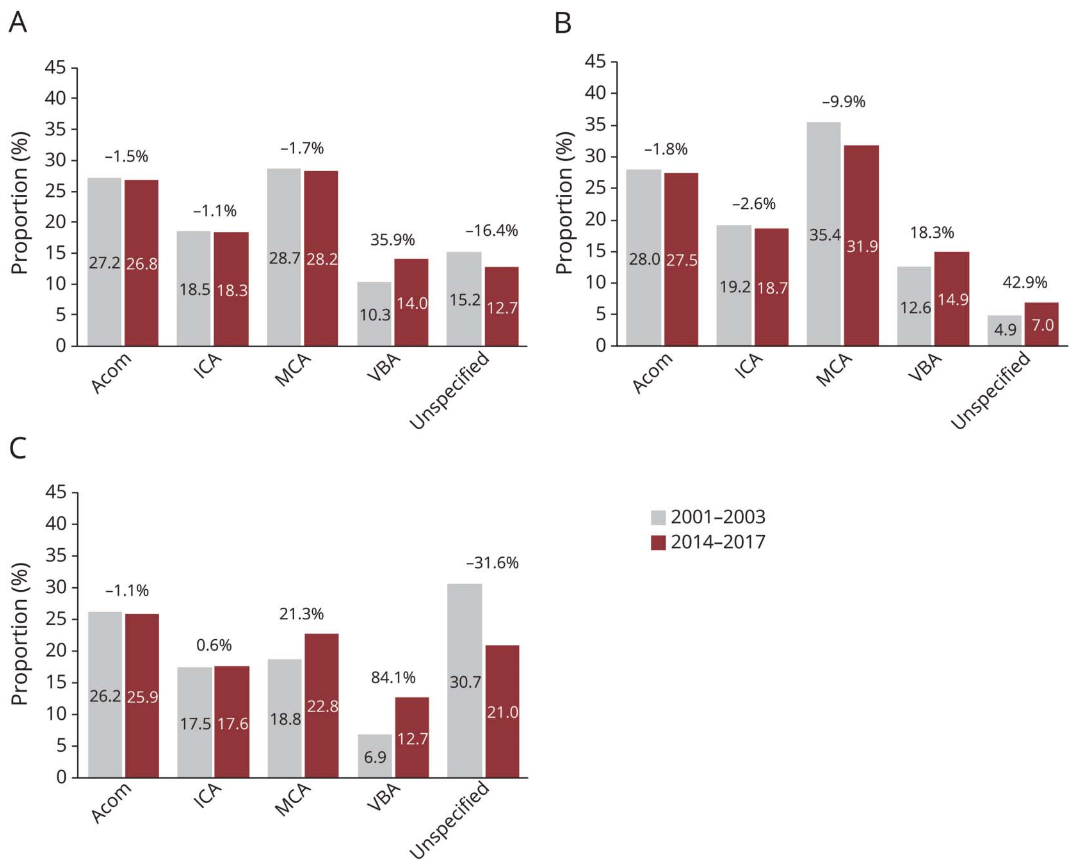
We identified 13,470 SAH cases (61.3% women) between 2001 and 2017. Of these, there were 3,581 (26.6%) Acoms, 2,480 (18.4%) ICAs, 3,970 (29.5%) MCAs, 1,552 (11.5%) VBAs, and 1,887 (14.0%) other/unspecified (I60.6–I60.9) RIA (Table 1). The most common RIA was MCA (34.8% of cases) in Finland and Acom (27.3% of cases) in New Zealand (Table 1). In New Zealand, European/others (26.7%) and Māori (29.7%) had the largest proportion of Acom RIAs, Pacific peoples had the most MCA RIAs (29.6%), and Asians had the highest proportion of ICA RIAs (24.7%). Between 2001–2003 and 2014–2017, the median age at SAH diagnosis increased from 56 to 60 years ($p < 0.001$). Individuals with anterior circulation RIAs were younger than those with VBA RIAs in both countries (57 vs 61 years, $p < 0.001$), and persons with other/unspecified RIAs were the oldest (63 years) (Table 1). The proportion of VBA RIAs did not differ by sex, whereas ICA RIAs were more common among women and Acom RIAs more common among men in both countries (Table 1). Between 2001–2003 and 2014–2017, the proportion of SAH that were VBA RIAs increased by 35.9% (from 10.3% to 14.0%) (Figure 1); this trend was evident in both countries, albeit more prominently in New Zealand (Figure 1). Over the same period, the proportion of other/unspecified RIAs increased by 42.9% in Finland (from 4.9% to 7.0%) but decreased by 31.6% in New Zealand (from 30.7%

Table 1 Baseline Characteristics of SAH Cases by RIA Location and Country

	No. of SAHs (% of total)	No. of hospitalized SAHs (% of total)	Median age, y (IQR)	No. of women (%)
Finland				
Anterior circulation	6,724 (79.9)	5,421 (83.0)	57 (49–68)	3,919 (58.3)
Acom	2,202 (26.2)	1,854 (28.4)	57 (48–66)	1,085 (49.3)
ICA	1,594 (18.9)	1,392 (21.3)	59 (50–70)	1,100 (69.0)
MCA	2,928 (34.8)	2,175 (33.3)	57 (49–68)	1,734 (59.2)
VBA	1,101 (13.1)	652 (10.0)	63 (53–73)	644 (58.5)
Other/unspecified	589 (7.0)	460 (7.0)	57 (49–67)	324 (55.0)
Overall	8,414 (100.0)	6,533 (100.0)	58 (49–68)	4,887 (58.1)
New Zealand				
Anterior circulation	3,307 (65.4)	3,083 (69.3)	55 (46–65)	2,237 (67.6)
Acom	1,379 (27.3)	1,302 (29.3)	54 (46–65)	786 (57.0)
ICA	886 (17.5)	842 (18.9)	57 (48–68)	700 (79.0)
MCA	1,042 (20.6)	939 (21.1)	54 (45–64)	751 (72.1)
VBA	451 (8.9)	388 (8.7)	56 (48–69)	282 (62.5)
Other/unspecified	1,298 (25.7)	975 (21.9)	67 (53–80)	844 (65.0)
Overall	5,056 (100.0)	4,446 (100.0)	57 (48–69)	3,363 (66.5)
Combined				
Anterior circulation	10,031 (74.5)	8,504 (77.5)	57 (48–67)	6,156 (61.4)
Acom	3,581 (26.6)	3,156 (28.8)	56 (47–65)	1,871 (52.3)
ICA	2,480 (18.4)	2,234 (20.4)	58 (49–70)	1,800 (72.6)
MCA	3,970 (29.5)	3,114 (28.4)	56 (48–67)	2,485 (62.6)
VBA	1,552 (11.5)	1,040 (9.5)	61 (51–72)	926 (59.7)
Other/unspecified	1,887 (14.0)	1,435 (13.1)	63 (51–76)	1,168 (61.9)
Overall	13,470 (100.0)	10,979 (100.0)	58 (48–69)	8,250 (61.3)

Abbreviations: Acom = anterior communicating artery; ICA = internal carotid artery; MCA = middle cerebral artery; RIA = ruptured intracranial aneurysm; SAH = subarachnoid hemorrhage; VBA = vertebrobasilar artery.
The ICD-10 codes represented by each RIA location: anterior circulation (I60.0–I60.3), Acom (I60.2), MCA (I60.1), ICA (I60.0 or I60.3), VBA (I60.4 or I60.5), and other/unspecified (I60.6–I60.9).

Figure 1 Changes in the Proportion of Each RIA Location Among All SAH Cases Between 2001–2003 and 2014–2017 in (A) Finland and New Zealand Combined, (B) Finland, and (C) New Zealand



Acom = anterior communicating artery; ICA = internal carotid artery including posterior communicating artery; MCA = middle cerebral artery; RIA = ruptured intracranial aneurysm; SAH = aneurysmal subarachnoid hemorrhage; VBA = vertebrobasilar artery.

to 21.0%) (Figure 1). The increase in the proportion of VBA RIAs was most prominent in men, older cases, hospitalized patients, and NZ ethnic groups of European/others and Māori (eFigures 3–6).

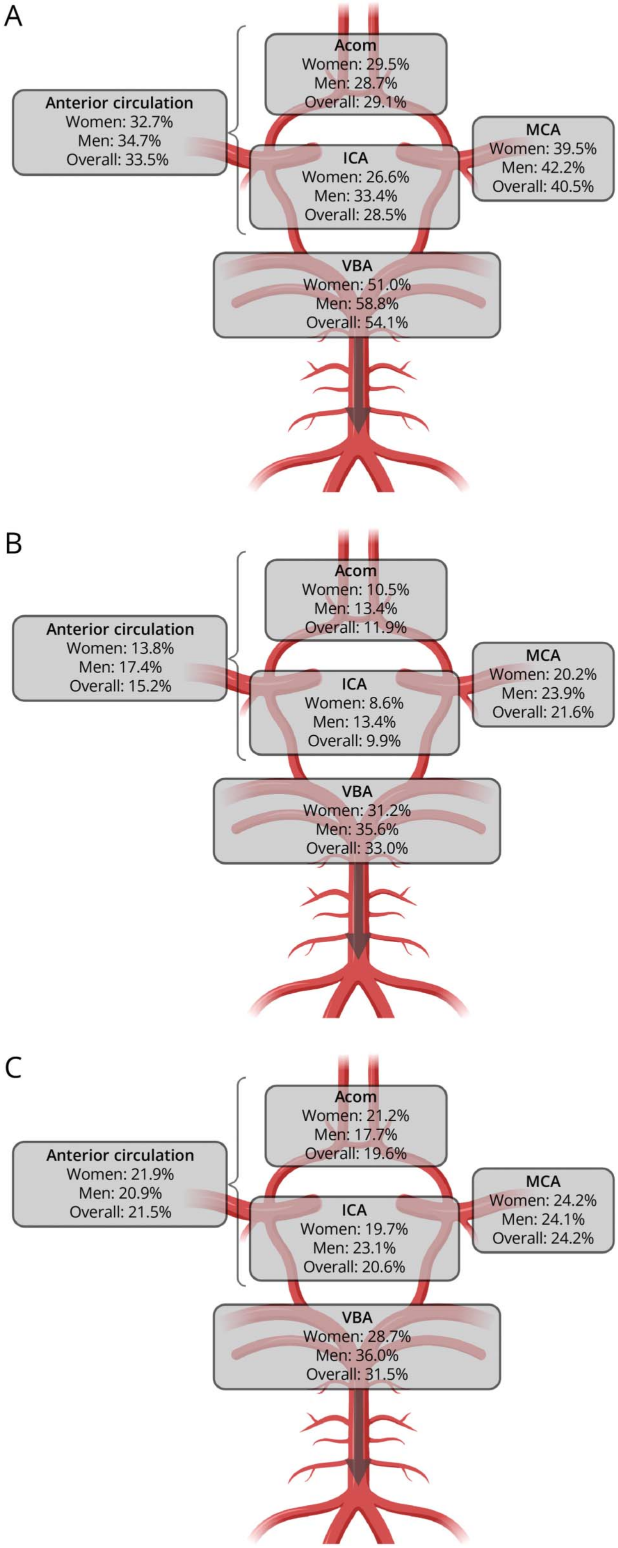
Case Fatality by RIA Location

After excluding 1,887 cases with other/unspecified RIA (I60.6–I60.9), the CFR analyses comprised 11,583 SAH cases (61.1% women) (Table 1). The crude overall 30-day CFRs (including prehospital deaths) were 29.1% for Acom, 28.5% for ICA, 40.5% for MCA, and 54.1% for VBA RIAs (Figure 2, eTable 10). Compared with MCA RIAs, the age-, sex-, and country-adjusted overall risk of SAH death within 30 days was 22% (95% CI 15%–29%) higher in VBA, 27% (22%–31%) lower in Acom, and 31% (26%–36%) lower in ICA RIAs (Table 2). The crude sudden death rates were 11.9% for Acom, 9.9% for ICA, 21.6% for MCA, and 33.0% for VBA RIAs (Figure 2, eTable 10). After adjustments, compared with MCA RIAs, sudden death rates were 42% (30%–55%) higher for VBA, 39% (32%–45%) lower for Acom, and 51% (44%–57%) lower for ICA RIAs (Table 2). The location-specific differences in 30-day CFRs of hospitalized patients were more modest (Table 2, Figure 2). These findings were broadly consistent in both countries and across subnational geographical regions (Table 3, eTables 11 and 12). The location-specific differences in overall 30-day CFRs and sudden death rates were mostly similar in women and men (Table 2, Figure 2) and were consistent across countries (eTables 13 and 14). The highest-to-lowest rankings of overall 30-day CFR and sudden death rates by RIA location (VBA > MCA > Acom/ICA) were identical in all age groups in both countries (Table 3), the exception being in younger NZ cases, where Acom RIAs had higher CFRs than MCA RIAs (eTables 13 and 14). By ethnicity, VBA RIAs had the highest overall 30-day CFR and proportion of sudden deaths among Europeans/others, Māori, and Pacific populations, whereas in Asian cases, VBA RIAs had the highest sudden death rates but the second highest overall 30-day CFR after the MCA RIAs (Table 3). The main results by ICD-10 codes and other/unspecified RIAs are presented in eTables 15 and 16.

Temporal Trends in CFR by RIA Location

From 2001–2003 to 2014–2017, the crude overall 30-day CFRs decreased from 60.5% to 48.9% in VBA RIAs, from 45.0% to 40.8% in MCA RIAs, and from 30.3% to 29.0% in Acom RIAs. In contrast, the 30-day CFR of ICA RIAs largely

Figure 2 Crude (A) Overall 30-Day CFR, (B) Proportion of Sudden-Death SAHs, and (C) 30-Day CFR in Hospitalized Patients With SAH by RIA Location in Finland and New Zealand (Combined)



The downward gray arrow indicates that VBA extends inferiorly beyond the gray box. Acom = anterior communicating artery; CFR = case fatality rate; ICA = internal carotid artery including posterior communicating artery; MCA = middle cerebral artery; RIA = ruptured intracranial aneurysm; SAH = aneurysmal subarachnoid hemorrhage; VBA = vertebrobasilar artery.

Table 2 RRs for SAH Death Between RIA Locations by Sex in Finland and New Zealand (Combined)

	Adjusted RR (95% CI)[a]		
	Proportion of sudden-death SAHs	30-d CFR in hospitalized patients with SAH	Overall 30-d CFR of SAH
Women			
Acom	0.57 (0.49–0.66)	0.72 (0.64–0.82)	0.74 (0.68–0.80)
ICA	0.44 (0.37–0.52)	0.79 (0.71–0.89)	0.65 (0.59–0.70)
MCA	Reference	Reference	Reference
VBA	1.40 (1.24–1.57)	1.06 (0.92–1.20)	1.18 (1.09–1.27)
Men			
Acom	0.64 (0.55–0.74)	0.74 (0.64–0.86)	0.73 (0.66–0.80)
ICA	0.58 (0.47–0.72)	0.89 (0.75–1.06)	0.79 (0.70–0.89)
MCA	Reference	Reference	Reference
VBA	1.46 (1.27–1.67)	1.30 (1.10–1.52)	1.29 (1.19–1.41)
Overall			
Acom	0.61 (0.55–0.68)	0.76 (0.69–0.83)	0.73 (0.69–0.78)
ICA	0.49 (0.43–0.56)	0.77 (0.70–0.85)	0.69 (0.64–0.74)
MCA	Reference	Reference	Reference
VBA	1.42 (1.30–1.55)	1.14 (1.03–1.27)	1.22 (1.15–1.29)

Abbreviations: Acom = anterior communicating artery; CFR = case fatality rate; ICA = internal carotid artery; MCA = middle cerebral artery; RIA = ruptured intracranial aneurysm; RR = risk ratio; SAH = subarachnoid hemorrhage; VBA = vertebrobasilar artery.
[a] Adjusted for age, sex, and country.

remained the same (Table 4). After adjusting for age, sex, and country, the overall 30-day CFRs decreased significantly by 24% (95% CI 13%–34%) in VBA RIAs and 15% (5%–24%) in MCA RIAs, decreased slightly in Acom RIAs (12% [−3% to 24%]), and remained stable in ICA RIAs (Table 4). These decreases were observed only in Finland, where the reductions in CFRs for VBA, MCA, and Acom RIAs were driven by decreasing sudden death rates (eTable 17). We found similarly declining sudden death rates for VBA and MCA RIAs among men, whereas in women, the decline was significantly greater for VBA RIAs than other locations (eTable 18). By age groups, the proportion of sudden deaths decreased most notably for VBA RIAs in young and older cases, whereas Acom RIAs showed the greatest decrease in middle-aged cases (eTable 19). The main results by specific ICD-10 codes and other/unspecified RIAs are also presented in eTables 20 and 21.

Discussion

We observed substantial differences in CFRs by RIA location, with VBA RIAs having the highest CFR being almost double the Acom and ICA RIAs. This location-specific CFR difference was more notable in prehospital sudden death rates than in hospitalized patients. Although the CFRs of VBA RIAs decreased mostly over time, their proportion of all SAHs increased by nearly 40% during the study period. These results emphasize that SAH is a heterogeneous disease where VBA and MCA RIA locations with the highest CFRs bear almost 70% of prehospital and 60% of all 30-day SAH deaths. Continuous improvements in hospital-based diagnostic and treatment modalities of SAH may have only had a modest effect on the CFRs of certain population groups and RIA locations, but it remains unclear whether improvements in symptom awareness and preventive and prehospital management strategies can decrease the persistently high CFRs of SAH. For example, an actionable strategy to prevent prehospital deaths could be to transport thunderclap headache patients with predefined high-risk features for aSAH directly to neurovascular centers, while directing lower-risk patients to local hospitals to prevent overwhelming tertiary services.

Reasons to explain why VBA RIAs have higher CFRs compared with anterior circulation RIAs include the proximity of VBA to the brainstem, an area responsible for various vital functions (e.g., ventilation, circulation, consciousness), which may contribute to rapid prehospital death. For example, acute neurogenic pulmonary edema (a serious complication seen most often in sudden-death SAHs[26,27]) appears substantially more common in VBA RIAs due to direct medullary compression by hematoma.[28] Similarly, posterior RIA location is associated with a rapid loss of consciousness.[29] Another suggested mechanism for sudden-death SAH is intraventricular

Table 3 Crude CFRs of RIA Locations Separately by Age, Ethnicity, and Country

	Proportion of sudden-death SAHs		30-d CFR in hospitalized patients with SAH		Overall 30-d CFR of SAH	
	No. of deaths/cases	Proportion (%) of sudden deaths	No. of deaths/cases	CFR (%)	No. of deaths/cases	CFR (%)
Young (<40 y)						
Anterior circulation	80/956	8.4	115/876	13.1	195/956	20.4
Acom	39/388	10.1	43/349	12.3	82/388	21.1
ICA	8/238	3.4	32/230	13.9	40/238	16.8
MCA	33/330	10.0	40/297	13.5	73/330	22.1
VBA	23/104	22.1	16/81	19.8	39/104	37.5
Middle-aged (40–64 y)						
Anterior circulation	785/6,105	12.9	929/5,320	17.5	1,714/6,105	28.1
Acom	235/2,245	10.5	321/2,010	16.0	556/2,245	24.8
ICA	120/1,389	8.6	200/1,269	15.8	320/1,389	23.0
MCA	430/2,471	17.4	408/2,041	20.0	838/2,471	33.9
VBA	213/811	26.3	142/598	23.8	355/811	43.8
Older (≥65 y)						
Anterior circulation	662/2,970	22.3	785/2,308	34.0	1,447/2,970	48.7
Acom	151/948	15.9	253/797	31.7	404/948	42.6
ICA	118/853	13.8	228/735	31.0	346/853	40.6
MCA	393/1,169	33.6	304/776	39.2	697/1,169	59.6
VBA	276/637	43.3	170/361	47.1	446/637	70.0
European/others[a]						
Anterior circulation	154/2,247	6.9	553/2,093	26.4	707/2,247	31.5
Acom	49/936	5.2	222/887	25.0	271/936	29.0
ICA	32/635	5.0	138/603	22.9	170/635	26.8
MCA	73/676	10.8	193/603	32.0	266/676	39.4
VBA	47/312	15.1	85/265	32.1	132/312	42.3
Māori[a]						
Anterior circulation	48/637	7.5	147/589	25.0	195/637	30.6
Acom	22/274	8.0	65/252	25.8	87/274	31.8
ICA	5/131	3.8	30/126	23.8	35/131	26.7
MCA	21/232	9.1	52/211	24.6	73/232	31.5
VBA	9/80	11.3	24/71	33.8	33/80	41.3
Pacific peoples						
Anterior circulation	10/174	5.8	39/164	23.8	49/174	28.2
Acom	3/69	4.4	16/66	24.2	19/69	27.5
ICA	3/34	8.8	7/31	22.6	10/34	29.4
MCA	4/71	5.6	16/67	23.9	20/71	28.2
VBA	4/23	17.4	5/19	26.3	9/23	39.1
Asian[a]						

Continued

Table 3 Crude CFRs of RIA Locations Separately by Age, Ethnicity, and Country *(continued)*

	Proportion of sudden-death SAHs		30-d CFR in hospitalized patients with SAH		Overall 30-d CFR of SAH	
	No. of deaths/cases	Proportion (%) of sudden deaths	No. of deaths/cases	CFR (%)	No. of deaths/cases	CFR (%)
Anterior circulation	12/183	6.6	41/171	24.0	53/183	29.0
Acom	3/68	4.4	12/65	18.5	15/68	22.1
ICA	4/69	5.8	16/65	24.6	20/69	29.0
MCA	5/46	10.9	13/41	31.7	18/46	39.1
VBA	3/26	11.5	7/23	30.4	10/26	38.5
Finland						
Anterior circulation	1,303/6,724	19.4	1,031/5,421	19.0	2,334/6,724	34.7
Acom	348/2,202	15.8	296/1,854	16.0	644/2,202	29.3
ICA	202/1,594	12.7	265/1,392	19.0	467/1,594	29.3
MCA	753/2,928	25.7	470/2,175	21.6	1,223/2,928	41.8
VBA	449/1,101	40.8	206/652	31.6	655/1,101	59.5
New Zealand						
Anterior circulation	224/3,307	6.8	798/3,083	25.9	1,022/3,307	30.9
Acom	77/1,379	5.6	321/1,302	24.7	398/1,379	28.9
ICA	44/886	5.0	195/842	23.2	239/886	27.0
MCA	103/1,042	9.9	282/939	30.0	385/1,042	37.0
VBA	63/451	14.0	122/388	31.4	185/451	41.0

Abbreviations: Acom = anterior communicating artery; CFR = case fatality rate; ICA = internal carotid artery; MCA = middle cerebral artery; SAH = subarachnoid hemorrhage; VBA = vertebrobasilar artery.
[a] Only New Zealand cases included.

hemorrhage (IVH) with rapid expansion of the fourth ventricle.[26,27] Consistent with our findings, the prevalence of IVH is reported to be higher in VBA than in anterior circulation RIAs.[30] Moreover, it is reported that persons with VBA RIAs tend to be more often smokers,[31] which predisposes to sudden-death SAH.[4] In hospitalized patients, the high CFR of VBA RIAs may be related to worse admission status,[7,11,32] higher rates of rebleeding,[33] and early hydrocephalus.[32] Regarding specific locations in the anterior circulation, the greater CFR of MCA RIAs may be related to higher rates of intracerebral hemorrhage (ICH)[17] and greater RIA size,[17,25] both of which are associated with poor outcomes.[8,9,17] The higher CFRs in VBA and MCA RIAs may also be due to the proportion of hospitalized patients who were treated in neurosurgical units being higher in Acom and ICA RIAs (eTable 22).

The declining prevalence of key risk factors for SAH death (i.e., smoking, hypertension[4]) in the general population,[3,14] alongside improved diagnostic resources (e.g., CT scan use[2]), better SAH awareness among physicians,[34] shorter treatment delays,[2] improved emergency medical management (e.g., more effective airway management[35]), and improved access to neurocritical care,[36] have all likely contributed to the overall decreasing trend in SAH CFRs and especially prehospital sudden deaths. Several factors may also have contributed to observed differences in the time-dependent trends in location-specific CFRs. For example, CFRs decreased significantly only in RIA locations more likely to cause sudden death (VBA and MCA), whereas RIA locations with a lower risk (Acom and ICA) did not show a similar decrease, suggesting that cases with VBA and MCA RIAs may have benefited more from improvements in emergency prehospital care.[2] Consistent with this, the proportion of hospitalized patients treated in neurosurgical units increased the most for VBA and MCA RIAs during the study period (eTable 23). The substantial decrease in CFRs of VBA RIAs may also be attributed to increased availability of endovascular treatment,[17] allowing more poor-grade VBA RIA cases that were previously managed conservatively to now receive operative treatment. In contrast, in a recent Finnish hospital-based study, CFRs decreased similarly in all RIA locations regardless of treatment modality.[17] The observation that CFRs decreased in Finland but not in New Zealand may relate to the observation that, in New Zealand, the proportion of other/

Table 4 Temporal Trends in Case Fatalities by RIA Location in Finland and New Zealand (Combined)

	Crude % (deaths/cases) in 2001–2003	Crude % (deaths/cases) in 2014–2017	Adjusted RR (95% CI)[a]	Adjusted annual decrease in % (95% CI)[a]
Proportion of sudden-death SAHs				
Anterior circulation	17.4 (323/1,862)	14.6 (310/2,119)	0.81 (0.70–0.93)	2.2 (1.3 to 3.2)
Acom	12.9 (88/681)	10.4 (80/773)	0.75 (0.57–1.00)	3.0 (1.2 to 4.8)
ICA	12.5 (58/463)	12.0 (65/542)	0.96 (0.70–1.32)	1.3 (−1.2 to 3.8)
MCA	24.7 (177/718)	20.5 (165/804)	0.79 (0.66–0.95)	2.0 (0.8 to 3.2)
VBA	39.9 (103/258)	25.1 (100/399)	0.61 (0.49–0.77)	3.8 (2.4 to 5.2)
30-d CFR in hospitalized patients with SAH				
Anterior circulation	22.3 (343/1,539)	22.5 (407/1,809)	0.93 (0.83–1.06)	0.4 (−0.4 to 1.2)
Acom	19.9 (118/593)	20.6 (143/693)	0.95 (0.77–1.17)	0.3 (−1.2 to 1.7)
ICA	19.5 (79/405)	21.2 (101/477)	1.06 (0.82–1.37)	−0.6 (−2.3 to 1.0)
MCA	27.0 (146/541)	25.5 (163/639)	0.85 (0.71–1.02)	1.2 (0.0 to 2.4)
VBA	34.2 (53/155)	31.8 (95/299)	0.80 (0.61–1.05)	1.6 (−0.2 to 3.3)
Overall 30-d CFR of SAH				
Anterior circulation	35.8 (666/1,862)	33.8 (717/2,119)	0.89 (0.82–0.97)	1.0 (0.5 to 1.6)
Acom	30.3 (206/681)	28.9 (223/773)	0.88 (0.76–1.03)	1.2 (0.2 to 2.2)
ICA	29.6 (137/463)	30.6 (166/542)	1.02 (0.85–1.22)	−0.1 (−1.3 to 1.2)
MCA	45.0 (323/718)	40.8 (328/804)	0.85 (0.76–0.95)	1.3 (0.6 to 2.1)
VBA	60.5 (156/258)	48.9 (195/399)	0.76 (0.66–0.87)	2.1 (1.3 to 3.0)

Abbreviations: Acom = anterior communicating artery; CFR = case fatality rate; ICA = internal carotid artery; MCA = middle cerebral artery; RIA = ruptured intracranial aneurysm; RR = risk ratio; SAH = subarachnoid hemorrhage; VBA = vertebrobasilar artery.
[a] Adjusted for age, sex, and country.

unspecified RIAs (i.e., SAH cases with notably high CFRs) decreased substantially from 31% to 21% during the study period, yet these cases were omitted from the CFR trend analyses. Indeed, a previous analysis based on the same nationwide NZ SAH cohort,[14] which included all SAHs regardless of location, showed a significant decrease in CFRs over the same period. The simultaneous decrease in the proportion of other/unspecified RIAs and the increase in VBA RIAs may suggest that diagnostics and availability of tertiary care for particularly VBA RIAs have improved. Moreover, given that some risk factors may predispose to RIAs at a particular location,[31] population-wide changes in risk factor profiles may have altered RIA location patterns. Taken together, high-quality, prospective, population-based studies are needed to identify factors driving aneurysm formation in high-risk locations (VBA and MCA), thus contributing to targeted prevention strategies to reduce the prevalence and CFR of such RIAs.

Our findings are consistent with historical population-based studies conducted over 40 years ago.[10,11] The limited evidence from these studies indicated higher sudden death rates in VBA than anterior circulation RIAs. More recent studies based on hospitalized patients have also shown higher CFRs for VBA[5-9] and MCA RIAs.[7,9] In addition, similar location-specific differences have been noted in both primary ICH and ischemic stroke, with infratentorial lesions and MCA/VBA occlusions showing the highest CFRs.[37,38] Moreover, we found that the proportion of VBA (13%) and MCA RIAs (34%) among localizable RIAs was higher in the current study cohort—which also included prehospital deaths—than reported in a pooled analysis of 7 hospital-based studies (11% and 18%, respectively).[6] Hence, when prehospital deaths are included, the actual proportion of VBA and MCA RIAs in the SAH population may be higher than previously thought. Finally, our findings are consistent with previous findings from a 20-year-long Finnish study, in which there was a significant increase in the proportion of posterior circulation RIAs in hospitalized patients.[12] However, we are not aware of previous population-based studies that have evaluated the location-specific CFR trends including prehospital SAH deaths.

Besides providing perhaps the most robust evidence on the prehospital and 30-day CFR of the most common RIA locations to date, a key strength of this study is its binational design. Finland and New Zealand are similar in size (~5 million residents) and health care infrastructure (both have high-quality universal public health care, comprehensive nationwide registers, and high autopsy rates in the general population[14,15]), yet they are in opposite hemispheres and differ in ethnic mix, geographic distribution, and vascular risk profiles. Thus, the observed similar results in both countries improve the generalizability of this study at least in countries with similar health care systems. However, this study has limitations. First, because we had no detailed data on SAH patients' risk factors (e.g., comorbidities and aneurysm size) or diagnostic and treatment methods (e.g., surgery and coiling), we were not able to investigate the underlying causes behind the differences in CFRs and their trends by RIA locations, nor adjust for them. For the same reason, we lacked data on prehospital deaths among patients with SAH with previously diagnosed, conservatively managed unruptured aneurysms, preventing the assessment of potentially avoidable deaths. Thus, future studies with more detailed individual-level data are needed to investigate whether location-specific SAH diagnostics, treatment, and prevention could be improved to decrease the burden of SAH. Second, although the external validation of our administrative dataset had good accuracy for the most common RIA locations, we were unable to distinguish less frequent locations without specific ICD-10 diagnosis codes in both anterior and posterior circulation territories. Indeed, approximately 14% of previously validated SAH cases did not have a location-specific coding (i.e., I60.0–I60.5) but were only defined as other/unspecified SAHs (i.e., I60.6–I60.9). According to the validation datasets, these excluded cases were predominantly RIAs in pericallosal and posterior inferior cerebellar arteries but also included many prehospital or conservatively treated deaths without autopsy or detailed diagnostics, so did not receive location-specific diagnosis (eTables 24 and 25). Consequently, excluding these cases may have led to underestimating the CFRs of RIA locations with high sudden death rates such as VBA and MCA. Third, because of the small number of verified cases in validation cohorts and the observation that many neurosurgical centers classify Pcom RIAs with the code I60.0, we further grouped the cases with noncommunicating ICA (I60.0) and Pcom (I60.3) RIAs into ICA category and cases with BA (I60.4) and VA (I60.5) RIAs into VBA category. Although this limited the number of included locations in our primary analyses, it created more robust estimates in presented categories that are also often used in clinical practices and SAH literature.[5,6,9] Furthermore, our sensitivity analyses showed that both BA (I60.4) and VA (I60.5) RIA locations had substantially higher CFRs, whereas noncommunicating ICA (I60.0) and Pcom (I60.3) RIA locations had substantially lower CFRs than the reference MCA location (eTable 15). Because the CFR trends were also similar within both grouped categories, the categorization had likely only a minor effect on the main findings and conclusions. Fourth, although the analyses were based on similar data collections and methodological approaches from Finland and New Zealand, case inclusion and validation procedures were not standardized. For example, because the size of the NZ validation cohort (ARCOS IV) was only one-eighth of the Finnish cohort (HUH Aneurysm Repair Register), the validation of certain locations such as VBA in NZ dataset included fewer than 10 patients (eTable 6). In contrast, we could not determine the location-specific PPVs for Finnish SAH cases, as the HUH Aneurysm Repair Register included only operatively treated cases while the administrative data collections did not have such data on the operative treatment. Nevertheless, because the overall findings were similar in both countries and compared with previously reported data from different hospitals outside of the validated regions (eTables 8), a significant systematic error is unlikely. Finally, our analyses were limited by heterogeneous study periods and by data ending in 2017. Consequently, further studies are required to evaluate how recent improvements in SAH management and prevention have shaped its CFR trends in the past decade. Despite these limitations, our approach provides superior population-based estimates on the SAH CFRs of the most common RIA locations in comparison to previous hospital-based studies in which over half of fatal SAHs (sudden deaths) were missing.[5]

SAH CFRs are substantially higher for VBA and MCA RIAs than for Acom and ICA RIAs, primarily due to differences in sudden death rates. Given the notable differences in location-specific SAH CFRs and temporal trends, future research incorporating more detailed individual-level data could help identify whether these high-risk locations could benefit from improvements in prehospital management strategies. These findings cannot be applied to the preventive treatment of unruptured aneurysms.

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Author Contributions

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